

Econ 272 / MGTECON 607: Intermediate Econometrics III – Section 8

May 28, 2025

Outline Today

1. Review
2. Practice Problems

1 Review of Concepts from Class

1.1 Questions

Based on the lecture, slides, and reading, you should be able to answer the following questions.

1.1.1 Lecture 16: Difference in Differences

1. What is the difference in difference estimator with two groups, two time periods?
2. What is the main assumption needed to identify causal effects with DiD? How does this rely on functional form?
3. What can we do to make the assumption more plausible?
4. What can we do to get standard errors?

1.1.2 Lecture 17: DiD with Heterogeneous Effects

- In what settings do TWFE and DiD yield the same results?
- What is the main problem with TWFE regression when there are multiple groups and time periods?
- What are alternative estimators that yield unbiased estimates?
- How do we choose between them?

1.1.3 Lecture 18: Synthetic Control

1. What is the main idea between synthetic control and its generalizations? Describe the method in words.
2. What do we mean by “vertical” and “horizontal” regressions?
3. What is the relation between TWFE and matrix completion?

4. Why do we need a regularization term for the matrix completion method?
5. What does the synthetic difference-in-differences estimator do and how does it compare to DiD and SC?
6. What do we mean by double robustness for the synthetic DiD estimator?
7. Under what conditions do the horizontal and vertical regressions yield the same point estimates?
8. What are some placebo analyses we can run to support the credibility of the synthetic control estimator?

1.2 Roadmap

We started our final chapter (the end is near!!) for the course, which is on panel data. After we have discussed several strategies the past three weeks on how to deal with data from observational studies in the cross-section setting, we are expanding our toolbox for methods that can deal with panel data. Panel data means that we observe a set of individuals over time. Specifically, we observe data for two or more groups in two or more periods. We assume that in some periods some groups are exposed to a new regime or policy that we wish to evaluate. This week we introduced the difference-in-differences estimator that relies on the so called parallel trends assumption, i.e. the treated group and the control groups' trends would have evolved similarly in absence of treatment. We also covered part of the recent literature on difference-in-differences with heterogeneous treatment effects, in which case two-way fixed effects is not equal to differences-in-differences anymore. There are several new estimators that were suggested - it is unclear how to choose between them and ideally they should all give you similar results.

We then introduced synthetic control methods (and talked about some recent extensions in the literature). Synthetic control methods provide a way of estimating the treatment effect on the treated, typically a single treated unit, by estimating the outcome of the treated unit in absence of treatment through construction of a synthetic treated unit. This synthetic unit is simply a convex combination of donor units, where the convex weights are chosen such that the synthetic unit mimics the outcome of the treated unit in absence of treatment (e.g. minimize pre-treatment error). We talked about the different types of regressions we can generally run in panel data settings, vertical vs horizontal regressions, and how they related to methods we introduced earlier in the quarter. We also introduced ways of thinking about inference in synthetic control (there is still no consensus on what to do) and extensions that are aimed at combining synthetic control and DiD as well as generalizing DiD more broadly using factor models.

1.3 Difference-in-Differences

In this section we introduce different panel data settings. Moreover, we introduce a first method, difference-in-differences, that exploits the specific structure of the data to evaluate treatment effects, where treatment is assumed to be given to some group(s) in some time period(s).

1.3.1 Setting

We can distinguish three different data configuration settings for panel data:

1. **Proper panel data:** we observe outcome Y_{it} for N units over T time periods, $i = 1, \dots, N$ and $t = 1, \dots, T$
2. **Repeated Cross-Sections:** we observe Y_i for units $i = 1, \dots, N$ in period T_i (different for different individuals), with individual i belonging to group G_i and G_i is binary indicating the treatment and control group
3. **Repeated Cross-Sections with Multiple Groups:** same as before, but now $G_i \in \{1, \dots, G\}$ that are split into treatment and control groups

1.3.2 Difference-in-Differences (DiD) Estimator

The DiD estimator is a widely universal and intuitive estimator. We will first go over in in the repeated cross-section setting and then in the proper panel data. In a naive way to get estimates of treatment effects, we could be (1) comparing the groups in the same point in time or (2) comparing the same group at different points in time. However, both of these approaches are potentially biased because of unobserved, systematic differences in the units or time periods. Under some additional assumptions that we will elaborate on below, we can compare the changes in a group that was exposed to the treatment in the second period, for example, to the change in the group that was never exposed to the new treatment and eliminate both biases. This is exactly what the difference-in-differences estimator does in a nutshell.

1.3.2.1 Repeated Cross-Section/Two Groups, Two Periods

Assume individual i belongs to a group, $G_i \in \{0, 1\}$ (where group 1 is the treatment group) and is observed in time period $T_i \in \{0, 1\}$. Letting the outcome be Y_i , the observed data are the triple (Y_i, G_i, T_i) .

We use the potential outcome notation from earlier in the class: $Y_i(0)$ is the outcome in absence of treatment and $Y_i(1)$ in presence of treatment. We denote the treatment indicator by $I_i = G_i T_i$. Thus, the realized (observed) outcome for individual i is

$$Y_i = (1 - I_i)Y_i(0) + I_i Y_i(1).$$

Table 1 depicts the setting.

		Period	
		before	after
Group	0	$\mathbb{E}[\bar{Y}_{00}]$	$\mathbb{E}[\bar{Y}_{01}]$
	1	$\mathbb{E}[\bar{Y}_{10}]$	$\mathbb{E}[\bar{Y}_{11}]$

Table 1: 2×2 Set-Up

Assumptions. In the standard DiD model, the outcome for individual i in the absence of intervention satisfies

$$Y_i(0) = \alpha + \beta T_i + \gamma G_i + \varepsilon_i,$$

where β denotes the time component and γ the group-specific, time-invariant component. ε_i denotes the errors term, i.e. the unobservable characteristics of the individual. It is independent of G_i and

T_i . To interpret the DiD treatment effect estimand, we need to make additional assumptions about how the outcomes are generated in presence of the intervention. Usually, we assume that the treatment effect is **constant** across individuals, so $Y_i(1) - Y_i(0) = \tau$. This assumption, in combination with the modeling assumption, yields the following model for the realized outcome

$$Y_i = \alpha + \beta T_i + \gamma G_i + \tau I_i + \varepsilon_i.$$

If the treatment effect differs across individuals, the standard DiD estimand gives the *average effect of the intervention on the treatment group*.

Note. An equivalent version of this is what is the well-known **parallel trends** assumption:

$$Y_{it}(0) = \alpha_i + \beta \mathbf{1}(\{time = post\}) \quad \text{or} \quad \mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|G_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0)|G_i = 0].$$

The parallel trends assumption is fairly strong and does not always hold in practice. Figure 1 visualizes this assumption for multiple time periods. Note that the parallel trends assumption is not

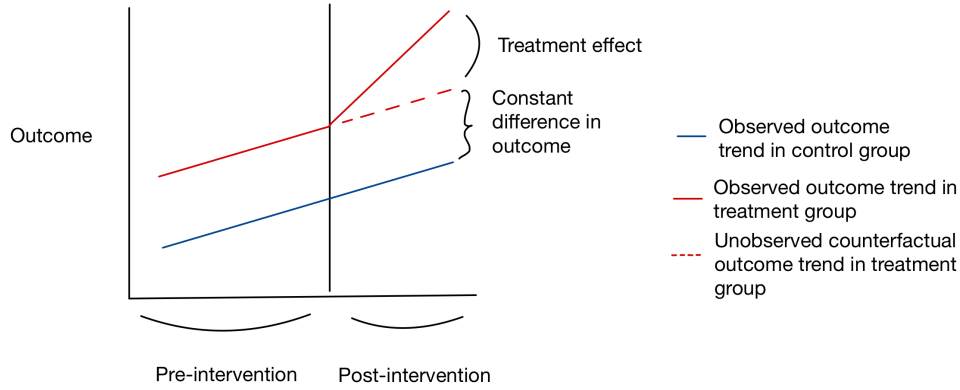


Figure 1: Parallel Trends Assumption

testable as it says something about the outcome evolution in absence of treatment *post-treatment*, which we don't observe. A common assessment for whether this assumption is intuitively valid is to check the trends pre-treatment. If they look parallel, we generally think we can trust this identification strategy.

Estimand. The standard DiD estimand is then given by

$$\begin{aligned} \tau^{did} = & \underbrace{\left[\mathbb{E}[Y_i|G_i = 1, T_i = 1] - \mathbb{E}[Y_i|G_i = 1, T_i = 0] \right]}_{\text{Difference 1}} \\ & - \underbrace{\left[\mathbb{E}[Y_i|G_i = 0, T_i = 1] - \mathbb{E}[Y_i|G_i = 0, T_i = 0] \right]}_{\text{Difference 2}} \end{aligned}$$

Estimation. An easy way to get an estimate for the estimand of interested is to simply replace all true expectations with their sample averages yielding

$$\hat{\tau}^{DiD} = \bar{Y}_{11} - \bar{Y}_{10} - (\bar{Y}_{01} - \bar{Y}_{00}),$$

where $\bar{Y}_{gt} = \frac{1}{N_{gt}} \sum_{i:G_i=g, T_i=t} Y_i$ is the average outcome in time period t and group g and $N_{gt} = \sum_{i=1}^N \mathbf{1}_{G_i=g, T_i=t}$ is the number of observations in time period t and group g .

Alternatively, we can use the specification of the outcome model to estimate the estimand by OLS:

$$\min_{\alpha, \beta, \gamma, \tau} \sum_{i=1}^N (Y_i - \alpha - \beta T_i - \gamma G_i - \tau I_i)^2.$$

This leads to the exact same estimate as before

$$\hat{\tau} = \bar{Y}_{11} - \bar{Y}_{10} - (\bar{Y}_{01} - \bar{Y}_{00}).$$

Variance and Variance Estimation. We can use pretty standard results to obtain the variance of the DiD estimator. If we assume homoskedastic error terms, we find

$$V(\tau)_{homosk} = \sigma^2 \left(\frac{1}{N_{11}} + \frac{1}{N_{10}} + \frac{1}{N_{01}} + \frac{1}{N_{00}} \right),$$

where we can estimate σ^2 by $S^2 = \frac{1}{N-1} \sum_{i,t} (Y_i - \bar{Y})^2$ to get an estimate of the variance.

The robust (White) variance under heteroskedasticity are

$$V(\tau)_{heterosk} = \frac{\sigma_{11}^2}{N_{11}} + \frac{\sigma_{10}^2}{N_{10}} + \frac{\sigma_{01}^2}{N_{01}} + \frac{\sigma_{00}^2}{N_{00}},$$

where the within group/time-period variance σ_{gt}^2 can be estimated by

$$S_{gt}^2 = \frac{1}{N_{gt} - 1} \sum_{i:G_i=g, T_i=t} (Y_i - \bar{Y}_{gt})^2$$

to get an estimate of the robust variance.

Covariates. Given the OLS representation for estimation, we can easily (and flexibly) incorporate exogenous/pre-treatment covariates in this model, namely

$$Y_i = \alpha + \beta T_i + \gamma G_i + \tau I_i + \delta' X_i + \varepsilon_i,$$

where ε_i is now also independent of the covariates in addition to the group/time dummies. We can again use OLS to estimate the parameters and estimate the variance in a similar way as before too. The covariates can also be interacted with the group and time dummies or the treatment indicator.

1.3.2.2 Panel Data

Suppose we now observe "proper" panel data, where we observe the same units at multiple points in time. For each individual $i = 1, \dots, N$, we observe $(G_i, Y_{i0}, Y_{i1}, W_{i0}, W_{i1})$ with G_i being the group indicator as before and W_{it} denoting the treatment value for unit i at time t .

Assumption. We specify the same model as in the repeated cross-section

$$Y_{it} = \alpha + \beta t + \gamma G_i + \tau W_{it} + \varepsilon_{it}.$$

This model is still justified by the parallel trends assumption.

Estimation. We can estimate the above model by OLS just as before. Without covariates, this estimation procedure leads to the exact same double difference as before in the two period case. We can alternatively also estimate it with the slightly adapted regression function

$$Y_{it} = \mu + \underbrace{\alpha_i}_{\text{fixed effect 1}} + \underbrace{\beta_t}_{\text{fixed effect 2}} + \tau W_{it} + \varepsilon_{it},$$

which we refer to as a **two-way-fixed-effect** regression (normalizing one of the α_i 's and one of the β_t 's to zero). Alternatively, we can also run the regression

$$\dot{Y}_{it} = \dot{W}_{it}\tau + \eta_{it},$$

where we doubly-demean the variables

$$\dot{Y}_{it} = Y_{it} - \frac{1}{T} \sum_{s=1}^T Y_{is} - \frac{1}{N} \sum_{j=1}^N Y_{jt} + \frac{1}{NT} \sum_{s=1}^T \sum_{j=1}^N Y_{js}$$

and similarly for $\dot{W}_{it}$.

1.3.2.3 Relation to Unconfoundedness

In the panel data settings, we could also apply our unconfoundedness approach based on lagged outcomes from a few weeks back. In this subsection, we will compare the two approaches. The overall take-away is that neither approach is right or wrong a priori, they both simply make pretty different assumptions and the researcher needs to choose which one is more plausible based on her substantive knowledge. Unconfoundedness given lagged outcomes would suggest the following equation

$$Y_{i1} - Y_{i0} = \alpha + \beta Y_{i0} + \tau G_i + \eta_i,$$

and the DiD assumption suggests

$$Y_{i1} - Y_{i0} = \alpha + \tau G_i + \varepsilon_i.$$

While it might seem that the unconfoundedness assumption is less restrictive by freeing up the coefficient on the lagged value of the outcome, the DiD assumption implies that adjusting for lagged outcomes in fact compromises the comparison because Y_{i0} may in fact be correlated with ε_i . In fact, in panel data settings, it may be less believable that Y_{i0} is uncorrelated with the error term, which would invalidate our unconfoundedness assumption. Hence, there are two contradicting forces here when we include Y_{i0} as a regressor in the regression. Overall, it is hard to evaluate the two assumptions ex ante without any context. Intuitively, when we are thinking about applying the unconfoundedness assumption, we assume that we find units that are close to the pre-treatment values of the treated units, which might seem more appealing and believable than thinking about comparing units that might be far apart pre-treatment. Moreover, for DiD assumptions to hold, the omitted variables are time-invariant and that might not seem plausible in many settings of interest.

Recommendation in practice: Since the data cannot tell us what approach to use, one answer is to check the robustness of your findings using alternative identifying assumptions. Ideally, you will find broadly similar results using both models. Moreover, there is actually a nice property about misspecification for the fixed effect model and the unconfoundedness model, namely you

can think of both as bounding your causal effect because if unconfoundedness is right, but you estimate a fixed effect regression, your estimates will be too big. Similarly, the other way around your estimates will be too small.

1.3.2.4 Multiple Groups and Multiple Time Periods

We can extend these methods to settings (repeated cross-section and panel data) for the case where we have multiple groups and multiple time periods. In these settings, issues like autocorrelation of the error terms might arise, leading to improper inference.

Model. We slightly update our model in this new setting

$$\begin{aligned} Y_i &= \beta_{T_i} + \gamma_{G_i} + \tau I_i + \varepsilon_i \\ &= \sum_{s=0}^S \beta_s \mathbf{1}_{T_i=s} + \sum_{g=0}^K \gamma_g \mathbf{1}_{G_i=g} + \tau I_i + \varepsilon_i, \end{aligned}$$

where we now observe $S + 1$ time periods and $K + 1$ groups. The error structure suggested by Bertrand et al. (2004) and Hansen (2006) in this setting is

$$\varepsilon_i = \eta_{G_i T_i} + \nu_i,$$

which allows for a permanent individual level error term ν_i as well as a group-specific time-varying component η_{gt} . The unit level error term has the following properties

$$\mathbb{E}[\nu_i \nu_j] = \begin{cases} \sigma_\nu^2 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

The second error component entails the autocorrelation structure, to be precise,

$$\eta_{gt} = \alpha \eta_{gt-1} + \omega_{gt},$$

where $\omega_{gt} \perp \eta_{gt-1}, \omega_{gt} \perp \omega_{gs}, s \neq t$ and $\mathbb{E}[\omega_{gt}^2] = \sigma_\omega^2$. Thus, we find the following structure for the second component

$$\mathbb{E}[\eta_{gt} \eta_{hs}] = \begin{cases} \sigma_\eta^2 & \text{if } g = h, t = s \\ \sigma_\eta^2 \alpha^{|t-s|} & \text{if } g = h, t \neq s \\ 0 & \text{if } g \neq h. \end{cases}$$

Combining both error term components, we find the overarching structure

$$\mathbb{E}[\varepsilon_i \varepsilon_j] = \begin{cases} \sigma_\nu^2 + \sigma_\eta^2 & \text{if } i = j \\ \sigma_\nu^2 \alpha^{|T_i - T_j|} & \text{if } i \neq j, G_i = G_j \\ 0 & \text{otherwise.} \end{cases}$$

Note. We can also write this in panel data notation. Consider

$$Y_{it} = \beta_t + \gamma_i + \tau W_{it} + \varepsilon_{it}.$$

We can write the various assumption about the error component correlation structure as a specification for the full covariance matrix

$$\Omega = \mathbb{E}[\varepsilon \varepsilon'].$$

After rewriting out regression as

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

the variance of the OLS estimator is

$$\mathbb{V}(\hat{\beta}_{OLS}) = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\Omega\mathbf{X})(\mathbf{X}'\mathbf{X}^{-1}).$$

Alternatively, the researcher can also run generalized least squares (GLS) to obtain a better variance

$$\mathbb{V}(\hat{\beta}_{GLS}) = (\mathbf{X}'\Omega\mathbf{X})^{-1}$$

1.3.3.1 Nonlinear Extensions: Changes-in-Changes The DiD model is not robust to transformations of the data, e.g. we estimate different treatment effects when we estimate the DiD model in levels vs logs. Both these regression specifications make slightly different assumptions about the independence of the residuals in the linear regressions, thus they cannot both be valid at the same time. Generally, picking the functional form is not guided by theory and the researcher does not know which one is true. The changes-in-changes approach is a generalization of the standard DiD model in several ways, including robustness to functional form specification.

1.3.3.2 Continuous Data

Assumptions. We let U_i denote the unobserved characteristics of individual i . We assume that we can model the outcome in absence of the intervention as

$$Y_i(0) = h(U_i, T_i),$$

with $h(u, t)$ strictly increasing in u . Moreover, we assume that the distribution of U_i is allowed to vary across groups, but not over time within groups, so

$$U_i \perp T_i | G_i.$$

Note. We can always get the standard DiD model back if we impose three additional assumptions: (1) additivity ($U_i \mathbb{E}[U_i | G_i] \perp G_i$), (2) single index model ($h(u, t) = \phi(u + \delta t)$ for a strictly increasing function $\phi(\cdot)$) and (3) an identity transformation ($\phi(\cdot)$ is the identity function). Overall, these assumptions would imply that

$$Y_{it} = U_i + \delta t,$$

which is exactly the standard DiD model.

Estimand. We assume that group $G_i = 1$ is treated at time $T_i = 1$. Our objective is still to find an estimate for the average treatment effect

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y(1) | G_i = 1, T_i = 1] - \mathbb{E}[Y(0) | G_i = 1, T_i = 1],$$

which boils down to these two expectations after some rearranging. We observe the first one and thus only need to infer the distribution of $Y(0) | G = 1, T_i = 1$ from all other distributions available, which is identified under our assumptions. Let $F_{Y(0),gt}(y)$ denote the distribution function of $Y(0)$ given $G = g$ and $T = t$. We give a visualization of the observed distribution functions below:

To estimate a treatment effect, we are looking for the unobserved cdf $F_{Y(0),11}(y)$, which we can rewrite in terms of the known distributions as follows

$$F_{Y(0),11}(y) = F_{Y(0),10}(F_{Y(0),00}^{-1}(F_{Y(0),01}(y))).$$

		T_i	
		0	1
G_i	0	$F_{Y(0),00}$	$F_{Y(0),01}$
	1	$F_{Y(0),10}$	$F_{Y(1),11}$

Consider the visualization in Figure 2. We will slowly walk through the estimation of $\hat{F}_{Y(0),11}$. We start off at the bottom panel and consider value y . We are trying to find $F_{Y(0),11}$, denoted by the red ? in the bottom panel. We follow the formula above, starting with the inner most transformation, $F_{Y(0),01}(y)$. This transformation of y leads us to number 2 in the top panel. The value is equal to q' . Then, next transformation is $F_{Y(0),00}^{-1}(q')$. We follow the red arrows and find that this transformation leads us to number 3, which gives us y . In the last step, we use this value in $F_{Y(0),10}(y)$, leading us to number 4 in the bottom panel, which is q . Finally, we follow the last arrow horizontally and found the corresponding value of the cdf of $Y(0)|G = 1, T = 1$ for y , which is q .

Estimation. We can simply plug in the empirical distribution functions for each of the three components on the right hand-side of the equation. The estimated cdf can then be used to calculate the mean of $Y(0)$ given $G = T = 1$ and, consequently, the average effect of the new treatment.

1.3.3 Discrete Data

In settings with discrete outcomes, the standard DiD model is not very attractive since it may violate the restriction that the expected value of a binary variable is between 0 and 1. The strict monotonicity assumption, on the other hand, is very restrictive as well. In this case, it is more appropriate to assume weak monotonicity, i.e.

$$h(u, t) \text{ is non-decreasing in } u.$$

To achieve identification, we need to impose an additional assumption, namely

$$U \perp G|Y, T.$$

This gives the following expression for $\mathbb{E}[Y(0)|G = 1, T = 1]$ in the binary case

$$\mathbb{E}[Y(0)|G = 1, T = 1] = \begin{cases} \frac{\mathbb{E}[Y(0)|G=0, T=1]}{\mathbb{E}[Y(0)|G=T=0]} \mathbb{E}[Y(0)|G = 1, T = 0] & \text{if } \frac{\mathbb{E}[Y(0)|G=0, T=1]}{\mathbb{E}[Y(0)|G=T=0]} \leq 1 \\ 1 - \frac{1 - \mathbb{E}[Y(0)|G=0, T=1]}{\mathbb{E}[Y(0)|G=T=0]} (1 - \mathbb{E}[Y(0)|G = 1, T = 0]) & \text{if } \frac{\mathbb{E}[Y(0)|G=0, T=1]}{\mathbb{E}[Y(0)|G=T=0]} > 1 \end{cases}$$

1.4 DiD with Heterogeneous Effects

In this section, we talk about the recent diff-in-diff literature, where econometricians have been raising issues about TWFE estimation with multiple groups/periods and the underlying assumptions we need to make. We are trying to shed light on where these issues are coming from and present three proposed alternative estimators by de CHaisemartin and d'Hautfoeuille (2020), Callaway and Sant'Anna (2021) and Sun and Abraham (2021).

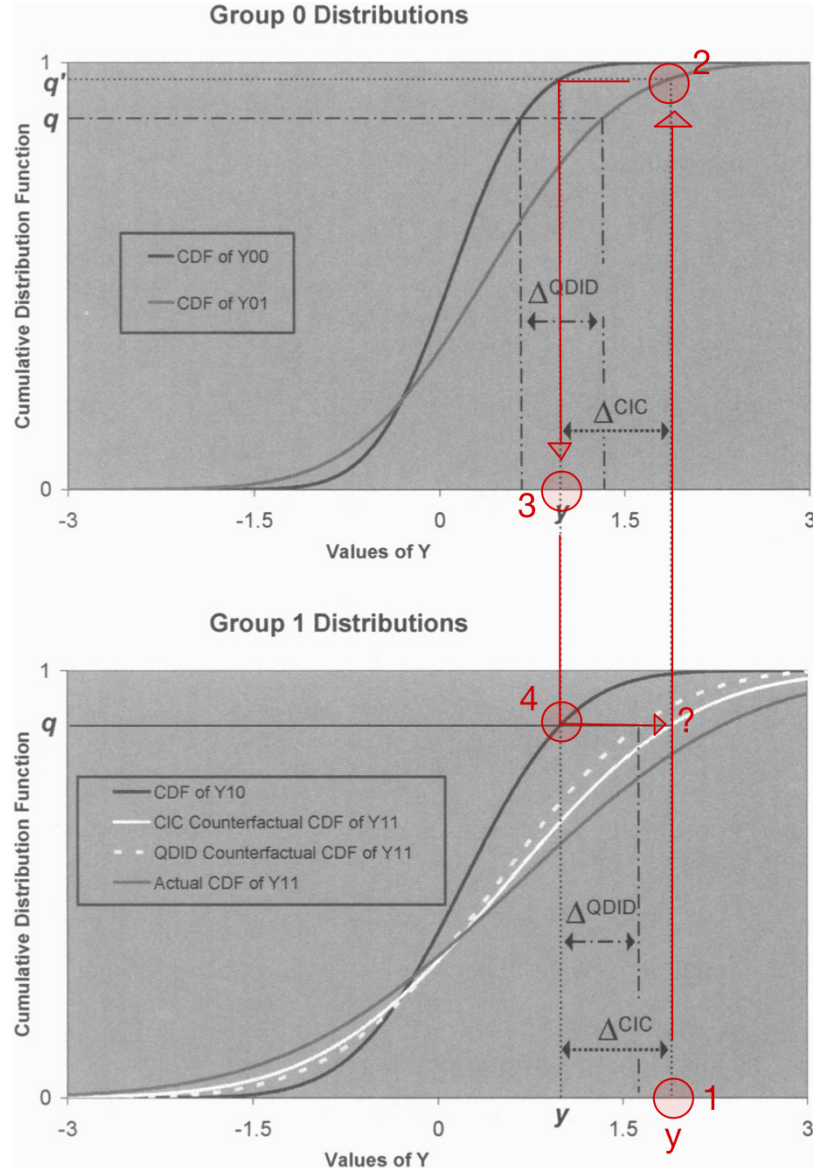


Figure 2: CiC Visualization

1.4.1 Set-Up

1.4.1.1 Notation

We are still dealing with the general panel data setting. For now, we introduce a few new pieces of notation:

- Z_i : time-invariant covariates (exogenous, pre-treatment)
- X_{it} : time-varying covariate (exogenous)

Note that the inclusion of time-varying covariates are in general somewhat controversial. Hence, most of the literature has ignored these.

1.4.1.2 Basic DiD and TWFE

Recall our definition for the standard DiD estimator for the easy two group/ two period example:

$$\hat{\tau}^{did} = (\bar{Y}^{post,treatment} - \bar{Y}^{post,control}) - (\bar{Y}^{pre,treatment} - \bar{Y}^{pre,control}).$$

Note that this is equivalent to our previous notation where we took the first difference to be the pre/post difference for the treated units and the second difference to be the pre/post difference for the control units. As we said previously, one of the differences is not enough as this might introduce bias due to the particular units being different because of unit specific effects or the time periods being different because of time specific effects.

We introduced an alternative characterization for $\hat{\tau}^{did}$ as the two-way-fixed-effect (TWFE) estimator $\hat{\tau}^{twfe}$ based on the following regression:

$$\min_{\alpha_i, \beta_t, \mu, \tau} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \mu - \alpha_i - \beta_t - \tau W_{it})^2,$$

with normalization on α_i , e.g. $\alpha_1 = 0$ or $\sum_i \alpha_i = 0$ and similar for β_t . A few notable differences to the simple DiD estimator is that this specification works for a general assignment mechanism W_{it} that does not split the data into a treatment and a control group as well as a pre- and a post-period (thus, two group/two period case).

Assumption. In the simple two group, two period design, we noted that both of these approaches yield unbiased estimates for the average treatment effect in the treatment group in the post-treatment period under the parallel trends assumption. In this lecture, we focus on the distinction and differences between DiD and TWFE in the multiple groups/time-periods setting. In general, we need an additional assumption for equivalence between these two approaches under this scenario, namely homogeneity of treatment effects across units and time periods (i.e. no dynamic changes). This is particularly important when we have a staggered adoption setting.¹ We are particularly looking at the interpretation of $\hat{\tau}^{TWFE}$ in staggered adoption settings as well as alternative estimators in these scenarios that do not suffer from the same issues. First, we define the staggered adoption setting in the next subsection.

1.4.1.3 Staggered Adoption

In terms of our treatment assignment matrix W_{it} , staggered adoption looks as follows:

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & 0 & \text{never adopter} \\ 0 & 0 & 0 & \dots & 0 & 1 & \text{late adopter} \\ 0 & 0 & 0 & \dots & 0 & 1 & \dots \\ 0 & 0 & 1 & \dots & 1 & 1 & \text{medium adopter} \\ 0 & 0 & 1 & \dots & 1 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 1 & 1 & \dots & 1 & 1 & \text{early adopter} \end{pmatrix}$$

In particular,

$$W_{it} \geq W_{it-1} \quad \forall t = 2, \dots, T, i = 1, \dots, N.$$

¹Applied researchers often conduct event studies instead of DiD designs given nature of the treatment (units adopt treatment at different time periods). These ones fall under the category of staggered adoption.

Denote the adoption date of a unit i by

$$A_i = \arg \min_t \mathbf{1}_{W_{it}=1}.$$

Never adopters have an adoption date of $A_i = \infty$. We can equivalently write our treatment indicator W_{it} now in terms of the adoption date, namely

$$W_{it} = \mathbf{1}_{t \geq A_i}.$$

Going forward, we view A_i as a (multi-valued) treatment. We slightly adapt our well-known potential outcomes notation to accommodate this:

- $Y_{it}(a)$: potential outcome for unit i in period t given adoption date a
- $Y_{it} = Y_{it}(A_i)$: realized outcome for unit i in period t
- $\bar{Y}_t(a) = \frac{1}{N} \sum_{i=1}^N Y_{it}(a)$: average potential outcome in period t for adoption date a
- $\bar{Y}_{ta} = \sum_i -i : A_i = a Y_{it} / \sum_{i:A_i=a} 1$: average realized outcome in period t for units with adoption date a
- $\dot{W}_{it} = W + it - \bar{W}_{\cdot t} - \bar{W}_i - \bar{\bar{W}}$: doubly-demeaned treatment indicator
- $V_W = \sum_{i,t} \dot{W} / NT$

1.4.2 The Interpretation of $\hat{\tau}^{TWFE}$ under Staggered Adoption

In general, a lot of the recent discussions in the DiD literature is concerned about the causal interpretation of $\hat{\tau}^{TWFE}$. We can always run a TWFE regression and get an estimate for $\hat{\tau}$, the question is whether we can causally interpret it or not. You can think about an analogous case with linear regression: we can always interpret the coefficient β as the best linear predictor, but we can only attach causal interpretation to it if we make additional assumptions.

1.4.2.1 Building Blocks

We start off by dissecting the bigger problem into smaller ones, specifically we are going to derive some building blocks. To start off, we think about the basic DiD estimator (two groups, two periods) for two different adoption dates, a' and a , and two different time periods s and t . In general, we will always compare a group that changed treatment status in the time periods of interest to a group that did not. Let

- $s < a \leq t$: group a adopts treatment between s and t
- $a' \leq s$ or $a' > t$: group a' adopts before s or group a' adopts after t

Note that the two scenarios in the second bullet point refer to the fact that the comparison group can either be true control units that have not adopted treatment yet or units that have adopted treatment but are not changing treatment status in this time frame. In terms of these adoption dates, we can then define the basic DiD estimator as

$$\hat{\tau}_{t,s}^{a,a'} = (\bar{Y}_{t,a} - \bar{Y}_{s,a}) - (\bar{Y}_{t,a'} - \bar{Y}_{s,a'}).$$

The overall treatment effect is simply a weighted combination of all these two group/two period comparisons. This means, there is no single treatment effect that we can recover - we will recover

different comparisons between units that we then aggregate to obtain a single estimate. Let us have a closer look at the simple building block to understand where issues arise with interpreting the TWFE estimator causally in light of only the parallel trends assumption. For now, we assume that the outcomes are generated according to the following model,

$$Y_{it}(0) = \mu + \alpha_i + \beta_t + \underbrace{\varepsilon_{it}}_{noise}.$$

For $Y_{it}(1)$, we assume that

$$Y_{it}(1) = Y_{it}(0) + \tau_{it},$$

that is, we let our treatment effects be unit and time-period specific, so we are interested in τ_{it} as our estimand.

We inspect the two separate cases from above separately:

Case I: $a' > t$

We find that

$$\begin{aligned}\bar{Y}_{a't} - \bar{Y}_{a's} &\approx \beta_t - \beta_s + noise \\ \bar{Y}_{at} - \bar{Y}_{as} &\approx \beta_t - \beta_s + \tau_{a,t} + noise\end{aligned}$$

Thus,

$$\hat{\tau}_{t,s}^{a,a'} = \tau_{a,t}.$$

Case II: $a \leq s$

We find that

$$\begin{aligned}\bar{Y}_{a't} - \bar{Y}_{a's} &\approx \beta_t + \tau_{a',t} - \beta_s - \tau_{a',s} + noise \\ \bar{Y}_{at} - \bar{Y}_{as} &\approx \beta_t - \beta_s + \tau_{a,t} + noise\end{aligned}$$

This means that we are using units in the control group that have already been treated. We find

$$\hat{\tau}_{t,s}^{a,a'} = \tau_{a,t} - (\tau_{a',t} - \tau_{a',s}).$$

Clearly, the second case does not give you the same result as the first case, *unless you assume constant treatment effects*, $\tau_{it} = \tau \forall i, t$. Here we can nicely see that, unless we assume constant treatment effects, we will not get an unbiased estimate of the treatment effect using the TWFE in the multiple units/time period example. In general, the bias is coming from the fact that we are using already treated unit as controls, thus people sometimes refer to this as "forbidden comparisons". Why is this a problem? In general, we could have that each individual treatment effect is positive, but if $\tau_{a',t}$ just dominates all other treatment effects, we can get a negative treatment effect overall.

Note. Dynamic treatment effects, e.g. increasing over time, could be one factor why we are worried about heterogeneity in treatment effects and will not believe in a constant treatment effect assumption.

1.4.2.2 Goodman-Bacon Decomposition

Goodman-Bacon propose a decomposition of τ^{TWFE} that characterizes the TWFE estimator in terms of three different groups:

$$\begin{aligned}\tau^{TWFE} = & \sum_{a \neq \infty} \omega^{a\infty} \hat{\tau}^{a,\infty} + \sum_{a \neq \infty, a' > a} \omega^{aa', a < a'} \tau_{a, a', a < a'} \\ & + \sum_{a \neq \infty, a' < a} \omega^{aa'} \hat{\tau}^{a, a', a' < a},\end{aligned}$$

with weights summing to 1, $\sum_{a \neq \infty} \omega^{a\infty} + \sum_{a \neq \infty, a' > a} \omega^{aa', a < a'} + \sum_{a \neq \infty, a' < a} \omega^{aa'} = 1$. In words, the TWFE is expressed as a convex combination of three different effects: (1) treatment effect for group that adopted treatment at time point a in comparison to a never-treated group, (2) treatment effect for group that adopted treatment at time point a to a group that adopted treatment at another time point a' , where a' is after a , thus we have "proper" control units and (3) same as (2), but we compare to a group a' , where a' is before a , thus comparing to units that have already been treated. Hence, this last term still creates a problem for interpretation as it can be negative in theory. This scenario especially arises if a big positive treatment effect for group a' in the period between a' and a relative to the earlier period could be misinterpreted as a trend.

1.4.3 Alternative Estimators

In the lecture, we discussed three alternative estimators in the lecture that yield causally interpretable estimators that relax the homogeneity in treatment effects assumption for TWFE: de Chaisemartin and d'Haultfoeuille (2020) (CH), Callaway and Sant'Anna (2021) (CS) and Sun and Abraham (2021) (SA).² We are going to talk through all of these in detail.

1.4.3.1 CH

CH consider a weighted combination of one-period ahead DiD estimators:

$$\hat{\tau}_{t,t-1}^{t,a} = (\bar{Y}_{t,t} - \bar{Y}_{t-1,t}) - (\bar{Y}_{t,a} - \bar{Y}_{t-1,a})$$

that lead to

$$\hat{\tau}_{+,t} = \frac{1}{T - (a - 1)} \sum_{a > t} \hat{\tau}_{t-1,t}^{t,a}.$$

To obtain the overall treatment effect, CH simply average over all the time periods post-treatment,

$$\hat{\tau}^{CH} = \sum_{t=2}^T \frac{N_t}{N} \hat{\tau}_{+,t}$$

where N_t is the number of adopters in period t . They rule out dynamic effects of the treatment and impose an additional parallel trends assumption on the outcome in presence of treatment as their proposed DiD estimator weighs a DiD estimator for units switching into treatment (compare those to untreated groups at both stages; usual DiD) and a DiD estimator for units switching out of treatment (compare those to treated groups at both stages). They thus allow for units switching

²See De Chaisemartin, C., & d'Haultfoeuille, X. (2022). Two-way fixed effects and differences-in-differences with heterogeneous treatment effects: A survey (No. w29691). National Bureau of Economic Research. for a nice and full discussion of these.

out of treatment.

1.4.3.2 CS

CS consider a comparison to a never-treated group only. Let

$$\hat{\tau}_{t,a-\delta-1}^{a,\infty} = (\bar{Y}_{t,a} - \bar{Y}_{a-\delta-1,a}) - (\bar{Y}_{t,\infty} - \bar{Y}_{a-\delta-1,\infty}),$$

where $\delta > 0$ indicates some anticipation effect. Moreover, consider

$$\hat{\tau}_{t,a-\delta-1}^{a,>t} = (\bar{Y}_{t,a} - \bar{Y}_{a-\delta-1,a}) - \frac{1}{T-t} \sum_{a'=t+1}^T (\bar{Y}_{t,a'} - \bar{Y}_{a-\delta-1,a'}).$$

In words, CS propose to compare the evolution of outcomes from $a-1$ to $a+l$ periods by comparing it to that of never treated. Overall, CS then consider several weighted averages of these, including the average effect of having been treated for $l+1$ periods across all groups reaching that treatment horizon. CS modify the parallel trends assumption on $Y_{at}(0)$ by a parallel trends assumption on the outcome without having ever been treated $Y_{at}(\mathbf{0}_t)$, more specifically, for $a \neq a'$

$$\mathbb{E}[Y_{a,t}(\mathbf{0}_t) - Y_{a,t-1}(\mathbf{0}_{t-1})] = \mathbb{E}[Y_{a',t}(\mathbf{0}_t) - Y_{a',t-1}(\mathbf{0}_{t-1})].$$

1.4.3.3 SA

SA consider the following quantity

$$\hat{\tau}_{t,a-1}^{a,\infty} = (\bar{Y}_{t,a} - \bar{Y}_{a-1,a}) - (\bar{Y}_{t,\infty} - \bar{Y}_{a-1,\infty}).$$

Their estimator is also a specific weighted combination of these, defined as follows

$$\hat{\tau}^{SA} = \sum_{t=2}^T \sum_{a=2}^T \hat{\tau}_{t,a-1}^{a,\infty} \cdot \frac{\Pr(A_i = a | 2 \leq A_i \leq T - (t - a))}{T - 1}.$$

Their proposed estimator is very similar to the one in CS as they also consider cohort and period specific effects that rely on the same parallel trends assumption as CS, but their weighting scheme is particularly focused on optimizing estimation precision. Moreover, their comparison allows to use the last-treated units as comparisons if there are no never treated units left. Lastly, SA approached the problem slightly differently as they looked at event studies instead of the usual DiD set-up and investigated TWFE in this setting.

1.4.3.4 How to Choose between These Estimators?

Given the many recently suggested estimators, you might be confused how to choose between them (or the wider class of estimators with non-negative weights on $\hat{\tau}_{t,s}^{a,a'}$). It is not entirely clear in practice which one to choose over the others as the advantages/disadvantages of one over the other are not yet well understood. A few tips to keep in mind when choosing:

- Constant additive fixed effects are more plausible in the short run $\rightarrow$ advocate for CH
- Never-treated groups may be qualitatively different from those treated at some point $\rightarrow$ advocate for CH
- One period ahead comparisons may miss long-run effects $\rightarrow$ SA, CS

In practice, ideally we would consider a whole class of weighted averages of these building blocks and look at how systematically different the weighting functions are, where the differences between the estimators are coming from and how sensitive they are to the violation of the assumptions or whether there is systematic variation of the treatment effect. In general, a good strategy on how to choose between these estimators is lacking. If you are lost about what estimators are out there and which setting they cover, see Figure 3 below for a summary from de Chaisemartin and d’Haultfoeuille (2022).³

1.5 Synthetic Control: Basics

According to Susan and Guido, Synthetic Control (SC) methods are ”arguably the most important innovation in the policy evaluation literature in the last 15 years.” (Athey and Imbens, 2017). As the quote suggests, we use synthetic control methods for policy evaluation. In the typical settings, we observe a small number of units (on an aggregate level, e.g. a country, state, etc.) out of which only one (or a few) units are treated. We observe the outcome data over time, thus these methods are used for panel data settings. Intuitively, you can think about SC methods as a data-driven method for comparative case studies. Instead of choosing a single unit to compare the treated unit to for policy evaluation, we choose a combination of similar units.

Example. There are a few canonical examples. Abadie and co-authors (2003, 2010, 2015) introduced the method in a series of papers ranging from terrorism in the Basque country in Spain to the tobacco control program (Proposition 99) in California over Germany’s reunification. All of these are macro-level interventions with outcomes observed on an aggregate level. Data is available for a small set of units and we only observe a single treated unit.

Contrast to DiD. There are a few differences to the DiD set-up. For DiD methods, we typically consider multiple units having received treatment and do not constrain ourselves to a settings with few units. Moreover, we usually assume that we have pretty granular data, e.g. individuals or cities, as opposed to states or countries. Lastly, both methods are justified using different assumptions. For SC, we often impose a latent factor structure which means we do not need to impose a parallel trends assumption. Figure 4 gives a high level overview of the comparison.

1.5.1 Set-Up

Just like in the DiD setting, we use synthetic control methods when we have panel data available. One difference is that we do require ”proper” panel data, i.e. we observe outcome Y_{it} for all $i = 1, \dots, N$ units over $t = 1, \dots, T$ time periods. We assume that we have $t = 1, \dots, T_0$ pre-treatment periods available and T_1 post-treatment periods. We observe a binary treatment indicator W_{it} . In matrix form,

$$\mathbf{W} = \begin{pmatrix} 1 & 1 & 0 & \dots & 1 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 1 & \dots & 0 \end{pmatrix}$$

We typically put some structure on this treatment assignment matrix (e.g. unit/time block structure, staggered adoption). In the SC case, we typically focus on a single treated unit (for a few

³See paper cited in earlier footnote for a sharper version of this table

Panel A: Estimators ruling out dynamic effects (outcome unaffected by past treatments)		
<i>Treatment</i>	<i>Estimators available</i>	<i>Stata¹⁵ commands</i>
Binary	de Chaisemartin and D'Haultfoeuille (2020)	did_multiplegt
	Imai and Kim (2018)	
	Borusyak et al. (2021)	did_imputation
Discrete	de Chaisemartin and D'Haultfoeuille (2020)	did_multiplegt
Continuous, with stayers	de Chaisemartin et al. (2022)	See paper
Continuous, w/o stayers	de Chaisemartin et al. (2022)	See paper
	Graham and Powell (2012)	gmm
	Chamberlain (1992)	gmm
Several treatments	de Chaisemartin and D'Haultfoeuille (2021b)	did_multiplegt
Panel B: Estimators allowing dynamic effects (outcome affected by past treatments)		
<i>Treatment</i>	<i>Estimators available</i>	<i>Stata¹⁵ commands</i>
Binary and staggered	Callaway and Sant'Anna (2021)	csdid
	Sun and Abraham (2021)	eventstudyinteract
	Borusyak et al. (2021)	did_imputation
	de Chaisemartin and D'Haultfoeuille (2021a)	did_multiplegt
Binary or discrete, non-staggered	de Chaisemartin and D'Haultfoeuille (2021a)	did_multiplegt
Continuous and staggered	de Chaisemartin and D'Haultfoeuille (2021a)	did_multiplegt
	Callaway et al. (2021)	
Continuous and non-stagg., with stayers	de Chaisemartin et al. (2022)	See paper
Continuous and non-stagg., w/o stayers	No estimator available yet	
Several treatments	de Chaisemartin and D'Haultfoeuille (2021b)	did_multiplegt

¹⁵All the Stata commands have R equivalents with the same name, except `eventstudyinteract` that does not have an R equivalent, and `csdid` whose R equivalent is called `did`.

Figure 3: A summary of DiD estimators allowing for heterogeneous treatment effects (time periods):

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

DiD and Synthetic Control for Panel Data Settings

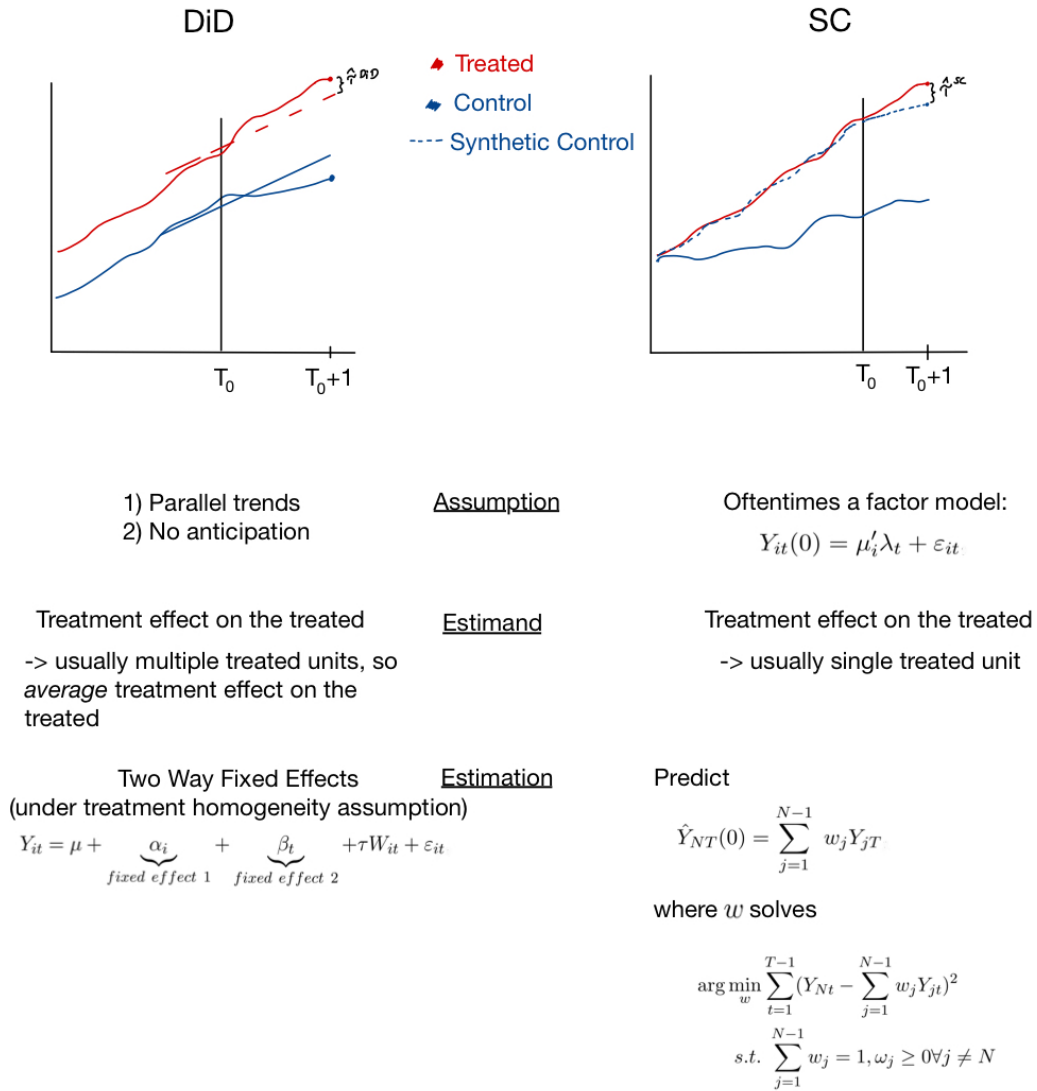


Figure 4: High-level comparison of DiD and SC

We also introduce potential outcome notation: $Y_{it}(0)$ for the outcome in absence of treatment and $Y_{it}(1)$ for the outcome in presence of treatment.

Estimand. The estimand that we are trying to recover with SC methods is the **treatment effect on the treated**,

$$\tau^{ATT} = \frac{\sum_{i,t} W_{it} (Y_{it}(1) - Y_{it}(0))}{\sum_{i,t} W_{it}}.$$

We could also focus on the treatment effect overall, but typically in these settings there are only a few treated units and, thus, the estimate on those is more interesting to look at or in a lot of cases also most policy relevant. Note that, if we assume there is only a single treated unit i and a single time period T , τ^{ATT} boils down to a single difference, namely

$$\tau^i = \underbrace{Y_{iT}(1)}_{=Y_{iT,observed}} - \underbrace{Y_{iT}(0)}_{not\ observed}.$$

Recall that we observe the first quantity because we are looking on the treatment effect *on the treated*, thus by definition we observe potential outcome $Y_{iT}(1)$ for it. Hence, we are left with a single challenge: predicting $Y_{iT}(0)$. There are three related literatures that find these predictions in a slightly different way (see Figure 5 for a visualization):

- **Vertical regression** (e.g. SC): $T - 1$ observations, $N - 1$ regressors
- **Horizontal regression** (e.g. causal literature with unconfoundedness): $N - 1$ observations, $T - 1$ regressors
- DiD and factor models

$$\mathbf{Y} = \begin{pmatrix} Y_{11} & Y_{12} & Y_{13} & \dots & Y_{1T} \\ Y_{21} & Y_{22} & Y_{23} & \dots & Y_{2T} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ Y_{N1} & Y_{N2} & Y_{N3} & \dots & Y_{NT} \end{pmatrix}$$

Figure 5: Visualization of horizontal and vertical regression: **Horizontal Regression** unit of observation, **Vertical Regression** unit of observation

Note that, for estimation purposes, we only have $N - 1$ units and $T - 1$ time periods available as we assume one unit to be treated and one time period to be treated, so we discard that data row/column.

Note. **Vertical regression** assumes a stable relationship across units over time following the idea that "similar units behave similarly". **Horizontal regression** assumes a stable relationship over time periods across units, following the idea that "history can explain everything". Typically, we use vertical regression if our data follows a "fat" matrix pattern (T large, N small) while we use horizontal regression if we have a "thin" matrix (T small, N large).

Recommendation in practice: N big, T small $\rightarrow$ matching makes sense. N small, T big $\rightarrow$ synthetic control makes sense

1.5.2 SC Method

Assumptions. The SC method is typically justified by a factor model. To be more precise, we assume that the movement of the outcome can be summarized by a few underlying, shared common factors across units. Mathematically,

$$Y_{it}(0) = \mu'_i \lambda_t + \varepsilon_{it},$$

where μ_i is a $F \times 1$ vector of factor loadings with F denoting the number of factors and λ_t a $F \times 1$ vector of factors and ε_{it} are transitory shocks. This is for the scenario without covariates. The original paper considers a scenario with covariates, with

$$Y_{it}(0) = \delta_t + \theta_t \mathbf{Z}_i + \mu_i' \lambda_t + \varepsilon_{it},$$

where δ_t is an unknown common factor with constant factor loadings across units (e.g. time fixed effect), $\mathbf{Z}_i$ is an $r \times 1$ vector of observed covariates that are not affected by the intervention and θ_t is a $1 \times r$ vector of unknown parameters.⁴

Estimation. In an ideal world, we would like to match units based on their (unobserved) factor loadings,

$$\sum_{j=1}^{N-1} \hat{w}_j \mu_j = \mu_i \text{ or at least } \approx,$$

where unit i is the treated unit. With the underlying factor structure, this would estimate a perfect counterfactual. In real life, we do not have access to the factors, so SC does the next best thing, it matches on pre-treatment period outcomes,

$$\sum_{j=1}^{N-1} \hat{w}_j Y_{jt} \approx Y_{it}, t = 1, \dots, T_0.$$

Abadie, Diamond and Hainmueller (2015) show that, under fairly standard conditions, the factor model implies that a synthetic control can fit $\mathbf{Z}_i$ and a long set of pre-intervention outcomes only as long as it fits $\mathbf{Z}_i$ and μ_i .

No covariates. In summary, SC methods estimate the unobserved counterfactual outcome, $\hat{Y}_{NT}(0)$ ⁵, by a convex combination of outcomes of other units that are similar to the treated unit (these units combined are usually referred to as a donor pool), i.e.

$$\hat{Y}_{NT}(0) = \sum_{j=1}^{N-1} w_j Y_{jT},$$

where w_j are estimated by optimizing pre-treatment fit, namely

$$\begin{aligned} \arg \min_w & \sum_{t=1}^{T-1} (Y_{Nt} - \sum_{j=1}^{N-1} w_j Y_{jt})^2 \\ \text{s.t.} & \sum_{j=1}^{N-1} w_j = 1, w_j \geq 0 \forall j \neq N \end{aligned}$$

Note that the convexity constraints on the weights ensure that the weight vector is sparse and they eliminate any extrapolation bias (since no extrapolation is allowed). Researchers oftentimes prefer

⁴There is no "nice" way of writing down the assumption of SC methods in terms of independence like the unconfoundedness, for example. The factor model assumption is a little bit of a roundabout way because if you are imposing a factor model, why not estimate it right away instead of using SC? However, this is still one of the most common ways to justify the method.

⁵Assume for the rest of these notes that, w.l.o.g., unit N is the treated unit and time period T is the treated time period.

sparse weights as well as positive weights for interpretability. In general, the weight restrictions serve two purposes: the restrictions make sense substantively and they implicitly regularize the regression when there are a lot of control units.

Covariates. The optimization problem above only takes into account lagged outcomes. In its original introduction, Abadie and co-authors include several covariates and only specific averages of the lagged outcomes. They introduce another weight matrix V that indicates how much weight is put on each covariate in the final optimization problem. The matrix V is estimated by cross-validation using some pre-treatment periods. These weights are then used in the final estimation of the "outer" weights w . Let the norm be $\|a\|_V = a'V^{-1}a$. Moreover, assume we standardize all of the covariates in Z_i . Mathematically, we define the following two optimization problems

Two-Part Procedure:

1. Outer Optimization:

$$\omega(V) = \arg \min_{\omega, \sum_{j=1}^{N-1} \omega_j = 1, \omega_j \geq 0 \forall j} \|Z_N - \sum_{j=1}^{N-1} \omega_j \cdot Z_j\|_V$$

2. Inner Optimization:

$$\hat{V} = \arg \min_V \|Y_{N,L} - \sum_{j=1}^{N-1} \omega_j(V) \cdot Y_{i,L}\|,$$

where V is positive semi-definite and $Y_{i,L}$ refers to the lagged outcomes for $t < T$.

Finally, following the two-part procedure, we find

$$\omega^* = \omega(\hat{V}) = \arg \min_{\omega, \sum_{j=1}^{N-1} \omega_j = 1, \omega_j \geq 0 \forall j} \|Z_N - \sum_{j=1}^{N-1} \omega_j \cdot Z_j\|_{\hat{V}}.$$

Note. If we include **all** lagged outcomes, this procedure will boil down to the case with **no covariates**. Since we optimize V based on pre-treatment outcomes, the optimal V would simply be the identity matrix on all lagged outcomes and every other covariates would get weight zero. V can sometimes be estimated based on only a subset of pre-treatment outcomes as well or simply chosen a priori by the researchers instead of this data-driven way.

Interpretation as Vertical Regression. An alternative characterization which draws the link to vertical regressions is given by Doudchenko and Imbens (2016). This relies on the knowledge that the solution for the ADH weights can be characterize in a single regression step in absence of covariates. Now, by freeing up the convexity constraints and allowing for an intercept, we can define the vertical regression (modified regression from original ADH) as

$$\hat{\tau}_{VR} = Y_{NT} - \hat{Y}_{NT}(0), \quad \hat{Y}_{NT}(0) = \alpha + \sum_{j=1}^{N-1} \omega_j Y_{jT}$$

where

$$(\alpha, \omega) = \arg \min_{\alpha, \omega} \sum_{t=1}^{T-1} (Y_{Nt} - \alpha - \sum_{j=1}^{N-1} \omega_j Y_{jt})^2 + \text{regularization term}.$$

Note that this formulation has **no** restrictions on w (thus, we allow for extrapolation), but it introduces an intercept and a regularization term. Regularization is important in this regression if N is large relative to T . Doudchenko and Imbens (2016) introduce an elastic net penalty to deal with the non-uniqueness of the weights if we are in this high-dimensional setting.

1.5.3 Unconfoundedness Methods/Horizontal Regression

The alternative way to obtain a prediction for $\hat{Y}_{NT}(0)$ is horizontal regression. Note that this approach relies on fundamentally different assumptions from the SC method. Generally, this method relies on unconfoundedness assumptions, e.g. unconfoundedness based on lagged outcomes. We spent a lot of time on unconfoundedness assumptions earlier in the quarter, so we are not going into details here. We can define the **horizontal regression** as

$$\hat{\tau}_{HR} = Y_{NT} - \hat{Y}_{NT}(0), \quad \hat{Y}_{NT}(0) = \alpha + \sum_{t=1}^{T-1} \lambda_t Y_{Nt}$$

where

$$(\alpha, \lambda) = \arg \min_{\alpha, \lambda} \sum_{j=1}^{N-1} (Y_{jT} - \alpha - \sum_{t=1}^{T-1} \lambda_t Y_{jt})^2 + \text{regularization term}.$$

Note that there are **no** restrictions on λ here either. However, once again, regularization will be important if $T \gg N$. These methods can also include matching, inverse propensity score weighty, doubly robust estimation or anything else we talked about earlier in the quarter.

1.5.4 Fixed Effects and Factor Models

We can view the TWFE model as a special case of a more general and flexible class of factor models with R factors, namely

$$Y_{it}(0) = \sum_{r=1}^R \gamma_{tr} \delta_{ir} + \varepsilon_{it}.$$

TWFE. Suppose $\gamma_{t1} = 1, \gamma_{t2} = \beta_2, \delta_{i1} = \alpha_i, \delta_{i2} = 1$. Then,

$$Y_{it}(0) = \sum_{r=1}^R \gamma_{tr} \delta_{ir} + \varepsilon_{it} = \gamma_{t1} \delta_{i1} + \gamma_{t2} \delta_{i2} + \varepsilon_{it} = \alpha_i + \beta_t + \varepsilon_{it}.$$

Note. If we choose $R = \min\{N, T\}$, we can fit the entire matrix exactly. This is both a feature and a bug of this representation. Thus, we are clearly worried about overfitting. Our goal is to find a matrix that allows for good out-of-sample predictions. There are two ways we can limit the number of factors: (1) fix R a priori or (2) regularize the objective function.

Consider the singular value decomposition (SVD) of the outcome matrix,

$$\mathbf{Y}_{N \times T} = \mathbf{U}_{N \times N} \mathbf{\Sigma}_{N \times T} \mathbf{V}_{T \times T}^T,$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthonormal matrices and $\mathbf{\Sigma}$ has non-zero elements (the singular values) only on the main diagonal. We limit the rank of the approximation by regularizing it as follows

$$\sum_{i,t} \left(Y_{it} - \sum_{r=1}^R \gamma_{tr} \delta_{ir} \right)^2 + \lambda \|\mathbf{\Gamma} \mathbf{\Delta}'\|_{nn},$$

where the nuclear norm of the approximation is the sum of the singular values and $\mathbf{\Gamma}$ is an $N \times R$ matrix capturing the time factors and $\mathbf{\Delta}$ a $T \times R$ matrix capturing the loadings. As usual with tuning parameters, we choose λ via cross-validation. Athey et al (2021) propose to explicitly include unit and time fixed effects (i.e. not penalize them) leading to

$$(\hat{\tau}^{FM}, \hat{\gamma}, \hat{\delta}) = \arg \min_{\gamma, \delta, \tau} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \alpha_i - \beta_t - \sum_{r=1}^R \gamma_{tr} \delta_{ir} - \tau W_{it})^2 + \text{regularization term}(\gamma, \delta).$$

This is known as matrix completion (MC) in the literature. It is not entirely sure where this fits in when we think about our thin and fat matrices. However, in simulations, it looks like it can deal very well with both.

1.5.5 How Do All These Methods Fit Together? ⁶

Given the different assumptions underlying all the panel data methods we have introduced (DiD, SC, MC), it might be surprising that we can combine them all under one framework: they can all be thought of as solving a least squares problem with some restrictions. We are walking through each method separately in the next subsections.

1.5.5.1 Synthetic Control

We can rewrite the SC objective function as

$$\hat{\tau}^{ADH} = \arg \min_{\tau, \gamma} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \gamma_t - \tau W_{it})^2 \times \omega_i^{ADH}.$$

Thus, SC (with the restrictions on the weights!) can be thought of as putting weights in the regression on different units, namely we only look at the units that are similar to the one we are interested in. In general, by just looking at the equation, this objective function looks not like a usual panel data objective function since there is not unit fixed effect. The differences between the units are all captured by reweighting. Under some conditions, we can base the standard errors on the regression interpretation taking the weights as given (even though they are estimated and depend on the outcome data).

1.5.5.2 Difference-in-Differences

Similarly, we can inspect the TWFE estimator and write

$$\hat{\tau}^{DiD} = \arg \min_{\tau, \gamma, \alpha} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \gamma_t - \alpha_i - \tau W_{it})^2.$$

This is a least squares regression with time and unit fixed effects, but no weights on the observations. It treats every unit as equally important and assumes that the differences between states are completely addressed by the unit fixed effect.

1.5.5.3 Synthetic Difference-in-Differences

Athey et al (2021) introduce synthetic difference-in-differences (SDiD) that combines the best of both worlds, SC and DiD. The SDiD estimator solves the following objective function

$$\hat{\tau}^{sdid} = \arg \min_{\tau, \gamma, \alpha} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \gamma_t - \alpha_i - \tau W_{it})^2 \times \omega_i^{ADH} \times \lambda_t^{ADH}.$$

⁶We still assume that unit N was treated at time period T .

This is a regression with unit and time fixed effects (like DiD) and with unit weights (like SC) and time weights. The unit weights are found in the same way as for SC, while the time weights satisfy

$$\lambda = \arg \min_{\lambda} \sum_{i=1}^N (Y_{iT} - \sum_{t=1}^{T-1} \lambda_t Y_{it})^2 + \text{regularization term}$$

$$s.t. \sum_{t=1}^{T-1} \lambda_t = 1 \quad \& \quad \lambda_t \geq 0$$

Note. This can easily be extended to the case where we have multiple post-treatment periods. In that scenario, we would try to match the average post-treatment period outcome.

In words, this is equivalent to down-weighting observations from distant past. This makes intuitive sense as you would think that more recent periods should be more predictive the post-treatment periods. Intuitively, the SDiD estimator runs a two step procedure. First, it reweights the control units to make the time trends of the synthetic unit and the treated unit parallel (though not identical as the SC estimator). Second, we run DiD on the reweighted panel, but only on the subset of time periods that are selected by $\hat{\lambda}_t$.

Figure 6 shows visualizations from the SDiD paper of the DiD, SC and SDiD estimators. I elaborated upon those in Figure 7 to include the pros and cons and the estimation procedure. The SDiD's time weights are given by the line at the very bottom of the SDiD figure (note that they are not on the same scale as the y -axis). They indicate that we are only using the last time periods for the DiD.

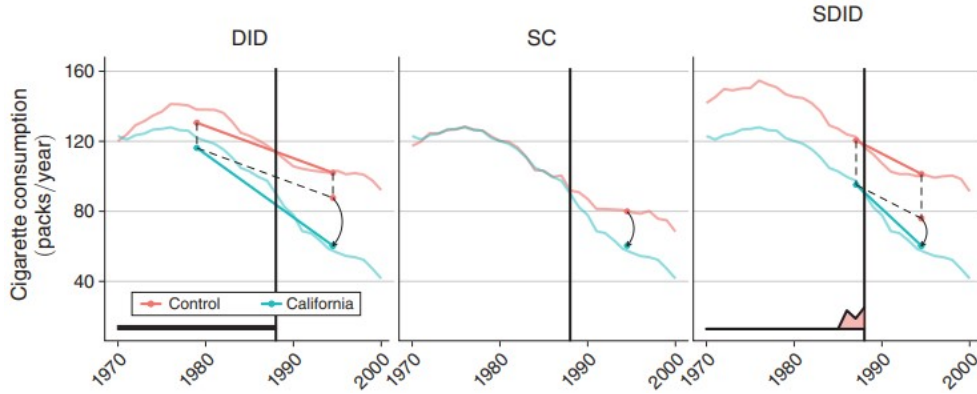


Figure 6: DiD, SC and SDiD

1.5.5.4 Factor Models

We can also rewrite our factor models in this framework, noting that those are a generalization of the SDiD model. The factor model estimator solves

$$\hat{\tau}^{FM} = \arg \min_{\alpha, \gamma, \tau} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \delta_i - \beta_t - \sum_{r=1}^R \gamma_{tr} \alpha_{ir} - \tau W_{it})^2 \times \omega_i^{ADH} \times \lambda_t^{ADH}.$$

Finding a Common Framework

- DiD

 - + invariant to additive unit-level shifts
 - + allows for valid large panel inference

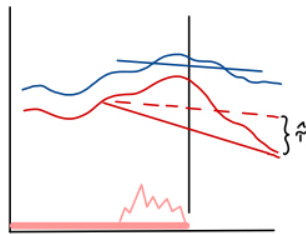
SC

 - + re-weights and matches pre-exposure trends to weaken parallel trends assumption



Synthetic DiD

$$\hat{\tau}^{sdid} = \arg \min_{\tau, \gamma, \alpha} \sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \gamma_t - \alpha_i - \tau W_{it})^2 \times \underbrace{\omega_i^{ADH}}_{\text{new: re-weights periods}} \times \lambda_t^{ADH}$$



Estimation procedure:

- 1) Reweigh units to make time trends parallel.
- 2) Apply DiD to the reweighted panel (only using a subset of the periods because of the time weights).

Figure 7: Finding a common framework: DiD, SC, SDiD

1.5.6 Properties of the Estimators

In this section, we compare the bias properties of the proposed estimators. We start with DiD. Recall that, when we talked about panel data, we needed to include a second difference to eliminate time and unit bias (that would arise from just comparing the treated unit post vs pre or just comparing the treated unit and control unit post treatment). Thus, DiD is a double bias reduction method:

$$\begin{aligned}\hat{Y}_{NT}^{did}(0) &= \frac{1}{(N-1)(T-1)} \sum_{i=1}^{N-1} \sum_{t=1}^{T-1} Y_{it} \\ &\quad + \frac{1}{N-1} \sum_{i=1}^{N-1} \left(Y_{iT} - \frac{1}{T-1} \sum_{t=1}^{T-1} Y_{it} \right) \quad (\text{time bias}) \\ &\quad + \frac{1}{T-1} \sum_{t=1}^{T-1} \left(Y_{Nt} - \frac{1}{N-1} \sum_{i=1}^{N-1} Y_{it} \right) \quad (\text{unit bias})\end{aligned}$$

SC, on the other hand, is a single bias reduction method, but with local weights:

$$\begin{aligned}\hat{Y}_{NT}^{ADH}(0) &= \frac{1}{T-1} \sum_{i=1}^{N-1} \sum_{t=1}^{T-1} \omega_i^{ADH} Y_{it} \\ &\quad + \sum_{i=1}^{N-1} \omega_i^{ADH} \left(Y_{iT} - \frac{1}{T-1} \sum_{t=1}^{T-1} Y_{it} \right) \quad (\text{time bias})\end{aligned}$$

SDiD combines both these properties and is a double bias reduction method with local weights for units and time:

$$\begin{aligned}\hat{Y}_{NT}^{sdid}(0) &= \frac{1}{(N-1)(T-1)} \sum_{i=1}^{N-1} \sum_{t=1}^{T-1} \omega_i^{ADH} \lambda_t^{ADH} Y_{it} \\ &\quad + \frac{1}{N-1} \sum_{i=1}^{N-1} \omega_i^{ADH} \left(Y_{iT} - \frac{1}{T-1} \sum_{t=1}^{T-1} \lambda_t^{ADH} Y_{it} \right) \quad (\text{time bias}) \\ &\quad + \frac{1}{T-1} \sum_{t=1}^{T-1} \lambda_t^{ADH} \left(Y_{Nt} - \frac{1}{N-1} \sum_{i=1}^{N-1} \omega_i^{ADH} Y_{it} \right) \quad (\text{unit bias})\end{aligned}$$

Thus, the SDiD estimator actually has the attractive property of **double robustness**:

- If a factor model holds, but the weights are good (e.g. ADH weights), SDiD is consistent.
- If the DiD model holds, but we use arbitrary weights, SDiD is consistent.

1.5.7 Relationship Between Horizontal and Vertical Regression

So far we have presented horizontal and vertical regression as two fundamentally different approaches to panel data that are justified under different assumptions. The earlier sections in these

notes showed how we can relate DiD and SC, also two methods with differing underlying assumptions. This section will focus on relating horizontal and vertical regression. The punch line is that Shen, Ding, Sekhon and Yu (2022) showed that there are circumstances where

$$\hat{\tau}_{VT} = \hat{\tau}_{HZ},$$

where *ADH* stands for the vertical regression and *UNC* for horizontal regression. In general, this does not hold if we add an intercept to the unconfoundedness regression or if we do not impose nonnegativity and additivity restrictions on the weights. To be more precise, we define the vertical and the horizontal regressions with slight modifications:

SC/Vertical Regression

$$\hat{\tau}_{ADH} = Y_{NT} - \hat{Y}_{NT}(0), \quad \hat{Y}_{NT}(0) = \sum_{i=1}^{N-1} \omega_i Y_{iT}$$

where

$$\omega = \lim_{\lambda \rightarrow 0} \arg \min_{\omega} \sum_{t=1}^{T-1} \left(Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it} \right)^2 + \lambda \|\omega\|_2$$

Unconfoundedness Methods/Horizontal Regression

$$\hat{\tau}_{UNC} = Y_{NT} - \hat{Y}_{NT}(0), \quad \hat{Y}_{NT}(0) = \sum_{t=1}^{T-1} \beta_t Y_{Tt}$$

where

$$\beta = \lim_{\lambda \rightarrow 0} \arg \min_{\beta} \sum_{i=1}^{N-1} \left(Y_{iT} - \sum_{t=1}^{T-1} \beta_t Y_{it} \right)^2 + \lambda \|\beta\|_2$$

Note. The penalty term in both regressions ensures uniqueness when the data is high-dimensional and even we find multiple exact solutions. In SC, we talked about how the additivity and non-negativity constraints act as a regularizer. Sometimes, if we are in a very very high-dimensional case, i.e. a lot of control units and only a few pre-treatment periods, one might think that those constraints are not enough to guarantee a unique solution. Hence, we could add this additional penalty term that penalizes the norm of the weights. Overall, this will guarantee a unique solution. Moreover, it is desirable to penalize the norm as this ensures that the weights are not too large which has important implications for the variance (e.g. the variance will be larger the larger the weights are).

Note 2. The penalty term for the horizontal regression is usually not imposed in practice, but you could imagine that you might be interested in doing that as a way to regularize the parameters in some way.

1.5.8 Inference in SC Settings ⁷

We have mentioned before that there is no consensus yet over how to get standard errors in SC settings. In general, when we are dealing with a single treated unit, this is a particularly challenging case because there is no central limit theorem for the average treated outcome, since we are not averaging. However, even if we have block assignment and multiple treated units, there are still multiple questions we have to answer to understand what uncertainty we would like to capture and choices we have to make for the estimand.

The first set of question deals with the latter. Are we looking for the variance of the average treatment effect, $ATE = \frac{1}{NT} \sum_{i,t} (Y_{it}(1) - Y_{it}(0))$, or the variance of the average treatment effect on the treated, $ATT = \sum_{i,t} W_{it} (Y_{it}(1) - Y_{it}(0)) / \sum_{i,t} W_{it}$? Or are we simply interested in testing $H_0 : Y_{it}(1) = Y_{it}(0) \forall i, t$? The second set of questions deals with what uncertainty we would like to capture. What is random? Are we assuming that (1) $Y_{it}(0) = mu_{it} + \varepsilon_{it}$, ε_{it} random, W_{it} fixed or (2) $Y_{it}(0)$ fixed and W_{it} random? What are sensible asymptotic sequences? Are we looking for more time periods, i.e. $T \rightarrow \infty$, more units $N \rightarrow \infty$ (Additional question: how? Does the distance between units get small? What properties should the additional units that we observe have?) or a combination of both?

In practice, people oftentimes report placebo analyses. Even here, we still need to decide whether we run the placebo analyses over units or time periods. In the next subsections, we are diving into details for both.

1.5.8.1 Placebo Analysis on Units

In this section, we describe the placebo analysis on units. The general idea is that we systematically loop through all untreated units, pretend that unit has been treated and construct an estimator based on that. Since we know the actual value (it is observed), we can use the squared difference to derive a variance estimator. To be more precise, for each unit $i \neq N$, we calculate an estimator for $\hat{Y}_{iT}$, using Y_{js} , $s = 1, \dots, T$, $j \neq i, N$. That is, we drop treated unit N completely from the data set and redo our SC analysis pretending unit i is the treated unit. The variance is then calculated by

$$\hat{V}^{unit \text{ placebo}} = \frac{1}{N-1} \sum_{j=1}^{N-1} (\hat{Y}_{jT} - Y_{jT})^2.$$

Figure 8 visualizes the procedure. We ignore the actually treated unit (bottom row) completely. For all other units, we loop through each of them individually, training the weights on the data to the left of the vertical black line (all pre-treatment periods) to infer the single value on the right (the boxed question mark).

1.5.8.2 Placebo Analysis on Time Periods

Similarly, we can conduct a placebo analysis on the time periods. The general idea here is that we systematically loop through a number of pre-treatment periods, pretend that period was the treatment period and construct an estimator for that period. Again, since we know the actual outcome value, we can use the squared difference to derive a variance estimator. More precisely, we calculate estimator $\hat{Y}_{Nt}$ for treated unit N in period $t \in \{T_0 + 1, \dots, T - 1\}$, using Y_{js} , $s = 1, \dots, t$, $j = 1, \dots, N$. T_0 here does not refer to the actual first treated period in real life, but to the

⁷Again, we assume unit N is treated in time period T .

$$\mathbf{Y} = \left(\begin{array}{cccc|c} Y_{11} & Y_{12} & Y_{13} & \dots & \boxed{?} \\ Y_{21} & Y_{22} & Y_{23} & \dots & Y_{2T} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \hline \cancel{Y_{N1}} & \cancel{Y_{N2}} & \cancel{Y_{N3}} & \dots & \cancel{Y_{NT}} \end{array} \right)$$

Figure 8: Unit Placebo Analysis

first "pretended" treatment period. We then calculate the variance as

$$\hat{V}^{time\ placebo} = \frac{1}{T - T_0 - 1} \sum_{t=T_0+1}^T -1(\hat{Y}_{Nt} - Y_{Nt})^2.$$

Figure 9 visualizes the procedure. We ignore the actually treated period (last column) and all periods up until t (determined as above) completely. Recall, period t is the period in which pretend treatment has happened. For all periods $t \in \{T_0 + 1, \dots, T - 1\}$, we loop through each of them individually, training the weights on the data to the left of the new vertical green line (all "pre-treatment" periods) to infer the single value on the right (the boxed question mark) for the treated unit.

$$\mathbf{Y} = \left(\begin{array}{ccc|ccc} Y_{11} & Y_{12} & Y_{13} & \dots & \cancel{Y_{1T}} \\ Y_{21} & Y_{22} & Y_{23} & \dots & \cancel{Y_{2T}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \hline Y_{N1} & Y_{N2} & \boxed{?} & \dots & \cancel{Y_{NT}} \end{array} \right)$$

Figure 9: Time Placebo Analysis

1.5.8.3 Interpretation

We start by making some (pretty strong) assumptions to make further progress on the interpretation of the placebo variances. First, we assume that the outcome follows a model with a systematic component, μ_{it} , and an idiosyncratic error term, ε_{it} , which is independent over time and units. Moreover, we assume that we can estimate the systematic component pretty accurately. Then, if $N, T \rightarrow \infty$,

$$\hat{Y}_{it} - Y_{it} \approx \varepsilon_{it}, \text{ with variance } \sigma_{it}^2.$$

In words, under these assumptions, we can approximate the idiosyncratic component ε_{it} by the difference between the predicted value $\hat{Y}_{it}$ and the true value Y_{it} . We assume that ε_{it} is mean zero and has variance σ_{it}^2 .

Under these assumptions, we can rewrite the placebo variances as

$$\hat{V}^{time\ placebo} \approx \frac{1}{T - T_0 - 1} \sum_{t=T_0+1}^{T-1} \varepsilon_{Nt}^2 \approx \frac{1}{T - T_0 - 1} \sum_{t=T_0+1}^{T-1} \sigma_{Nt}^2$$

$$\hat{V}^{unit\ placebo} \approx \frac{1}{N - 1} \sum_{j=1}^{N-1} \varepsilon_{iT}^2 \approx \frac{1}{N - 1} \sum_{j=1}^{N-1} \sigma_{iT}^2.$$

If we assume that there is heteroskedasticity in the variances over time and across units, e.g.

$$\sigma_{it}^2 = \sigma^2 V_i U_t, \quad V_i \sim \chi^2(K_N)/K_N, U_t \sim \chi^2(K_T)/K_T,$$

then

$$\hat{V}^{time\ placebo} \approx V_N \sigma^2$$

$$\hat{V}^{unit\ placebo} \approx U_T \sigma^2$$

Thus, both are valid on average, but they will be different in any given data set. Generally, if there is a lot of heteroskedasticity across units, we would want to use the time placebo variance as this gets us the variance for this specific unit. If, on the other hand, there is a lot of heteroskedasticity over time, we would want to use the unit placebo variance as this gets us the variance for the specific time period.

2 Practice Problems

2.1 Diff-in-Diff, Synthetic Controls (Practice Final Exam 2016, #2)

Suppose Texas introduces an income tax. We want to estimate the effect of the tax using data from two repeated cross sections of individuals from Texas, one before and one after the introduction of the income tax, and data for the same periods for individuals from other states that never had an income tax and that did not introduce one.

- Describe the Difference-In-Differences method for estimating the effect of the tax on Texas residents.
- Describe the Abadie-Diamond-Hainmueller synthetic control method for this question.
- Discuss the relative merits of the two methods, and describe when you would use one rather than the other.

Solution Sketch:

(a) Difference-in-Differences

- A = Difference in means of Texas: after minus before
- B = Difference in means elsewhere: after minus before
- $\hat{\tau}^{dd} = A - B$
- Could also describe it as diff between Texas and elsewhere after, minus diff between Texas and elsewhere before.

(b) Synthetic Control Method

- Aggregate units to get state averages to get "proper" panel data
- Find weights on other states such that weighted outcome of other states approximates weighted outcome in Texas pre-treatment as well as covariates are balanced, subject to constraints on the weights.
- Then estimate counterfactual Texas in the post-treatment as the weighted average of other states, using weights estimated above.
- Treatment effect estimate is difference between actual and counterfactual; could subtract the mean difference between actual and counterfactual from the pre-period.
- Doesn't actually work in two-period setup without covariates that is described here. ⁸

(c) **Discussion** Different answers can make valid points, there is no single solution.

- Many people prefer the SC method estimates when there are enough pre-periods, ideally to estimate on first half, verify on second half. Easy visual check whether weights "work."
- But could do the same with DD if we had multiple periods, i.e. plot difference between "to be treated" states and control states for each period, and hope that the difference is stable.
- SC appears to choose smarter weights – if appropriate, it can weight all controls equally, like DD does.
- But inference (s.e., confidence intervals) not (yet) well-developed for SC.
- In practice, probably just do both.

2.2 Diff-in-Diff, Synthetic Controls (Final Exam 2019, #2)

In 2018 California started requiring internet companies to withhold taxes on sales to California customers even if the company did not have a physical presence in California. Suppose you are an internet company that was affected by this, and you want to assess how much that cost you in terms of sales. You have data on California sales in 2018 and for twenty previous years, and data for twenty five other states that are not requiring online companies to collect this tax, both in 2018 and for the same twenty previous years.

- (a) Describe a difference-in-difference strategy to estimate this effect.
- (b) Describe how you could proceed assuming unconfoundedness of the treatment (the requirement to collect the taxes).
- (c) Describe a synthetic control strategy.
- (d) Which of these strategies would you prefer in this case (not considering other strategies!).

Solution Sketch:

⁸*New:* Technically, given the constraints on the weight vector, you do get a solution back even though we are only estimating it effectively on a single data point. However, you may not want to trust this relationship very much because it is only based on a single time period and, moreover, if you had chosen the donor units wisely, your treated unit lies in the convex hull and you will most likely not get a unique solution back, which the original ADH estimator cannot deal with.

(a) **Difference-in-Differences**

- Use two-way-fixed-effects regression
 - Treated unit: CA
 - Treated period: 2018
 - Pre-treatment periods: previous 20 years
 - Control group: 25 other states
- Or simple difference in means estimator where we average over all blocks of units/time periods
- Model still based on parallel trends

(b) We first state the unconfoundedness assumption:

$$W_i \perp (Y_i(0), Y_i(1)) | X_i.$$

In this setting, we find that

- W_i : requirement to collect taxes
- $Y_i(0)$: sales in California in 2018 in absence of the tax requirement
- $Y_i(1)$: sales in California in 2018 in presence of the tax requirement
- X_i : sales data from the past 20 years (lagged outcomes)

In words, given the sales data from past years, sales in 2018 in absence and in presence of the tax requirement are independent of the requirement to collect taxes. Phrased differently, there is no selection into the tax requirement, after adjusting for past sales outcomes.

Strategies to proceed to estimate the average treatment effect assuming unconfoundedness:

- **Matching:** We could use a matching estimator. This might be slightly awkward because we only have a single treated unit available (thus every control unit would have this treated unit as a match and the estimator might be very biased). (*We didn't cover this this year.*)
- **Regression:** We could alternatively also run a regression of Y_i on W_i and X_i and we can interpret the coefficient on W_i as our estimate for τ^{ATE} .

Note. We cannot do propensity score matching as there is only a single treated unit and we can thus not learn the propensity function.

(c)

- If you have individual sales data, aggregate that up to an aggregate quantity for each state (e.g. average sales in each state).
- Find weights on other states such that weighted outcome of other states approximates the weighted outcome in California pre-treatment, subject to convexity constraints on the weights (no covariates available according to prompt, so we can ignore those)
- Estimate the counterfactual sales in CA in the post-treatment as the weighted average of other states, using the weights as estimated above
- The treatment effect is the difference between the actual outcome and the estimated counterfactual one

(d) *There is not a single solution as each of the strategies has its own advantages/disadvantages. Make sure you explain why you are advocating for a particular method.*

The set-up lends itself nicely for an analysis using SC. We observe a single treated (aggregate) unit as well as data on 25 other states (control units) and for 21 periods (20 pre-treatment, one post-treatment). The SC method allows us to make full use of all the pre-treatment data available and not impose a restrictive parallel trends assumption like the DiD. Moreover, if DiD is the optimal weighting scheme, SC can always give us back DiD and only improve in terms of minimizing pre-treatment error (this is a separate point from thinking about whether the identification assumptions underlying one methods are more reasonable than for the other method). You might be worried about underlying differences between the treated state and the control states that are not captured by the pre-treatment sales, e.g. demographics, that would invalidate your unconfoundedness assumption.

2.3 Recent DiD (Final Exam 2023, #4)

Consider the two-way-fixed-effect estimator in a setting with staggered adoption.

- (a) What concerns have been raised in the case there is heterogeneity in the treatment effects?
- (b) Describe one solution to this that has been proposed.

Solution Sketch:

(a) The issue that the recent literature on DiD raises is that $DiD \neq TWFE$. In general, we cannot interpret the TWFE estimator causally unless we assume homogeneity of the treatment effects in addition to the well-known parallel trends assumption. This concern mainly arises from the fact that we compare units to already treated units (also known as a "forbidden comparison") since the difference between these includes the treatment effects themselves. If we assume that the treatment effect are heterogeneous across groups and/or time periods, those do not cancel out and we obtain a biased estimate of our estimand of interest.

(b) There are three estimators we talked about that have been proposed to address this solution:

- CH: use one-period ahead estimators only, their estimator allows for treatment switchers, but not dynamic effects
- CS: compare units only to never-treated units, their estimator does not allow for treatment switchers, but can capture dynamic effects; never-treated groups might be substantially different from those who are treated at some point
- SA: compare units to never-treated units just like CS, but weigh them optimally, their estimator does not allow for treatment switchers, but can capture dynamic effects; never-treated groups might be substantially different from those who are treated at some point

2.4 Synthetic Controls, (Final Exam 2022, #3)

Consider the Abadie-Diamond-Hainmueller (ADH) paper on synthetic control methods where they estimate the effect of the anti-smoking legislation.

- (a) Describe the ADH synthetic control method.

- (b) Discuss what happens if you include in the vector of variables that you try to match on in the first stage you include all the lagged outcomes.
- (c) What are the advantages and disadvantages of restricting the Synthetic Control weights to be nonnegative?
- (d) Would you recommend using synthetic control methods instead of difference in differences with the Card-Krueger minimum wage data?
- (e) Describe the synthetic difference in differences method.

Solution Sketch:

a) Assuming for simplicity that only country $i = N$ is treated in period $t = T$, SC imputes the counterfactual untreated outcome $Y_{NT}(0)$ as:

$$\hat{Y}_{NT}(0) = \sum_{j=1}^{N-1} w_j(\hat{V}) Y_{jT}$$

where the weight vector $w(V) \in \mathbb{R}^{N-1}$ is the minimizer of

$$\begin{aligned} w(V) = & \arg \min_w \left\| Z_N - \sum_{i=1}^{N-1} w_i Z_i \right\|_V \\ \text{s.t.} \quad & w_i \geq 0 \\ & \sum_{i=1}^{N-1} w_i = 1 \end{aligned}$$

where the vector $Z_i = [X_i, \dots, Y_{it}, \dots]$ contains relevant covariates X_i of country i as well as a subset of pre-treatment outcomes, e.g. $Y_{it} : t \in S$ for some $t \in S \not\equiv T$. The optimal weighting matrix $\hat{V}$ for the vector Z_i is found via a cross-validation approach, minimizing the prediction error for the remaining pre-treatment outcomes that were not included in Z_i :

$$\hat{V} = \arg \min_V \sum_{t \notin S, t < T} \left(Y_{N,t} - \sum_{i=1}^{N-1} w_i(V) Y_{i,t} \right)^2$$

The final treatment effect is given by:

$$\hat{\tau} = Y_{NT} - \hat{Y}_{NT}(0)$$

b) The problem with including all pre-treatment periods into Z_i , i.e. setting $S = \{1, \dots, T-1\}$, is that we are unable to estimate $\hat{V}$ using the cross-validation approach described above - the set $\{t \notin S, t < T\}$ is empty. Including pre-treatment outcomes in both steps would cause these elements of Z_i to receive all of the weight, which is unlikely to yield robust estimates in practice. Instead, one should carefully think about which pretreatment periods are most important to get right (usually the ones closer to T) and optimize $\hat{V}$ so we do well on those, and include the remaining pre-treatment periods into Z_i .

c) A key advantage of restricting the weights to be nonnegative is that it acts as a regularizer: we're estimating an $N-1$ -dimensional weight vector $w(V)$ as well as a potentially large weighting matrix V , and restricting the model to represent country N as a convex combination of the remaining countries removes a lot of degrees of freedom and thus susceptibility to random noise.

Fortunately, this restriction is fine for most countries, as intuitively most countries are "somewhere in between" other countries, by definition. However, again by definition, not all countries can be "somewhere in between" other countries, and so this restriction introduces a bias for countries at the extremes: to allow for extrapolation, we need to allow for weights which are negative. Even then, we likely want to restrict the weights to be small in absolute terms to limit the amount of extrapolation, e.g. by imposing an L2 penalty.

d) The Card-Krueger minimum wage data only has one pre-treatment and one post-treatment period for the treated and untreated states, so we cannot apply Synthetic Controls at the state level, because this would require multiple pre-treatment periods. DiD is thus a more natural choice. In contrast to SC however, this requires the parallel trends assumption, which we cannot verify based on the two periods available in the data. Since Card-Krueger observe multiple individuals in the two states however, one could apply the Changes-In-Changes estimator to weaken this assumption.

(e) Synthetic DiD estimates

$$\hat{\tau}^{SDID} = \arg \min_{\tau, \gamma, \alpha} \left(\sum_{i=1}^N \sum_{t=1}^T (Y_{it} - \gamma_t - \alpha_i - \tau W_{it}) \hat{w}_i \hat{\lambda}_t \right)$$

which can be viewed as generalizing Synthetic Controls and Difference-in-Differences, each of which are obtained as particular choices for $\hat{w}$ and $\hat{\lambda}_t$: SC arises with $\hat{w}_i = \hat{w}_i^{ADH}$, $\hat{\lambda}_t = \frac{1}{T}$ and DID arises with $\hat{w}_i = \frac{1}{N}$ $\hat{\lambda}_t = \frac{1}{T}$. SDID estimates both the time and unit weights analogously to ADH:

$$\begin{aligned} \hat{w} &= \arg \min_w \sum_{t < T} (Y_{Nt} - \sum_{i < N} w_i Y_{it})^2 \\ \hat{\lambda} &= \arg \min_{\lambda} \sum_{i < N} (Y_{iT} - \sum_{t < T} \lambda_t Y_{it})^2 \end{aligned}$$

In words: the synthetic DiD estimator solves a weighted least squares problem that weighs each time period and unit according to weights chosen in a similar fashion to SC. In a first step, the estimator reweights all control states such that the time trends of the synthetic unit and the treated unit are parallel (though not necessarily overlapping). In a second step, it then applies DiD to the reweighted panel, using only the subset of units that have positive weights $\hat{\lambda}_t$.

2.5 Synthetic Controls (Final Exam 2023, #3)

Consider the Abadie-Diamond-Hainmueller (ADH) paper on synthetic control methods where they estimate the effect of the anti-smoking legislation.

- Describe the ADH synthetic control method.
- Show that in the case with a single treated unit (say unit N) and a single treated period (period T) and no covariates, the ADH estimator can be written as a least squares regression estimator based on the specification

$$Y_{Nt} = \sum_{i=1}^{N-1} \omega_i Y_{it} + \tau W_{Nt} + \varepsilon_t,$$

subject to the simplex constraints on the weights ω_i .

- Discuss how you may use a placebo analysis to support the credibility of the synthetic control estimator.

Solution Sketch:

(a) Assuming for simplicity that only country $i = N$ is treated in period $t = T$, SC imputes the counterfactual untreated outcome $Y_{NT}(0)$ as:

$$\hat{Y}_{NT}(0) = \sum_{j=1}^{N-1} w_j(\hat{V}) Y_{jT}$$

where the weight vector $w(V) \in \mathbb{R}^{N-1}$ is the minimizer of

$$\begin{aligned} w(V) = & \arg \min_w \left\| Z_N - \sum_{i=1}^{N-1} w_i Z_i \right\|_V \\ \text{s.t.} \quad & w_i \geq 0 \\ & \sum_{i=1}^{N-1} w_i = 1 \end{aligned}$$

where the vector $Z_i = [X_i, \dots, Y_{it}, \dots]$ contains relevant covariates X_i of country i as well as a subset of pre-treatment outcomes, e.g. $Y_{it} : t \in S$ for some $t \in S \not\equiv T$. The optimal weighting matrix $\hat{V}$ for the vector Z_i is found via a cross-validation approach, minimizing the prediction error for the remaining pre-treatment outcomes that were not included in Z_i :

$$\hat{V} = \arg \min_V \sum_{t \notin S, t < T} \left(Y_{N,t} - \sum_{i=1}^{N-1} w_i(V) Y_{i,t} \right)^2$$

The final treatment effect is given by:

$$\hat{\tau} = Y_{NT} - \hat{Y}_{NT}(0)$$

In words: the SC estimator finds weights on other states such that the weighted outcome of other states approximates weighted outcome in California pre-treatment as well as covariates are balanced, subject to constraints on the weights. The weights for the different covariates are determined in an inner optimization procedure that uses the pre-treatment periods to evaluate how much weight to give to each covariate in the overall optimization procedure. Then we estimate counterfactual California in the post-treatment as the weighted average of other states, using weights estimated above. The treatment effect estimate is difference between actual and counterfactual.

(b) First, note that in the case with no covariates, the ADH estimator simply boils down to minimizing the squared pre-treatment error subject to the ADH constraints, namely

$$\hat{\omega} = \arg \min_{\omega, \sum_{i=1}^{N-1} \omega_i = 1, \omega_i \geq 0} \sum_{t=1}^{T-1} \left(Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it} \right)^2$$

and then we find

$$\hat{\tau}^{ADH} = Y_{NT} - \sum_{i=1}^N \hat{\omega}_i Y_{iT}.$$

Next, for LS, we aim to minimize the sum of squared residuals, thus

$$\min_{\tau, \omega} \sum_{t=1}^T \varepsilon_{Nt} = \sum_{t=1}^T (Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it} - \tau W_{Nt})^2,$$

where W_{Nt} is a vector of all 0s except for the very last entry at $W_{NT} = 1$. Using the Frisch-Waugh theorem, we find that

$$\begin{aligned} \hat{\tau} &= (W'_{Nt} W_{Nt})^{-1} W'_{Nt} Y_{Nt} - (W'_{Nt} W_{Nt})^{-1} W'_{Nt} [\mathbf{Y}_1 \cdots \mathbf{Y}_{N-1}] \omega \\ &= Y_{NT} - [Y_{1T} \ Y_{2T} \ \cdots \ Y_{N-1T}] \omega \\ &= Y_{NT} - \sum_{i=1}^{N-1} \omega_i Y_{iT} \end{aligned}$$

Thus, we find that the regression estimator and the SC estimator have the same solution form. It remains to show that we also get the same weights ω_i . We go back to our original optimization problems for each estimator, derive FOCs and show that the FOCs are the same for both problems.

SC no covariates:

$$\mathcal{L} = \sum_{t=1}^{T-1} (Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it})^2 - \lambda_1 \left(\sum_{i=1}^{N-1} \omega_i - 1 \right) - \sum_{i=1}^{N-1} \lambda_{2_i} w_i$$

Thus, we find for our FOCs that

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \lambda_1} &= \sum_i \omega_i - 1 = 0 \\ &\iff \sum_{i=1}^{N-1} \omega_i = 1 \\ \frac{\partial \mathcal{L}}{\partial \omega_i} &= \sum_{t=1}^{T-1} (Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it}) \cdot 2 \cdot (-Y_{it}) - \lambda_1 - \lambda_{2_i} = 0 \\ &\iff \sum_{t=1}^{T-1} -2Y_{it}(Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it}) - \lambda_1 - \lambda_{2_i} = 0 \end{aligned}$$

Regression:

$$\begin{aligned} \mathcal{L} &= \sum_{t=1}^T (Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it} - \tau W_{Nt})^2 - \lambda_1 \left(\sum_{i=1}^{N-1} \omega_i - 1 \right) - \sum_{i=1}^{N-1} \lambda_{2_i} w_i \\ &= \sum_{t=1}^T \left\{ (Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it})^2 - 2(Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it})\tau W_{Nt} + \tau^2 W_{Nt} \right\} - \lambda_1 \left(\sum_{i=1}^{N-1} \omega_i - 1 \right) - \sum_{i=1}^{N-1} \lambda_{2_i} w_i \end{aligned}$$

Thus, we find for our FOCs that

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \lambda_1} &= \sum_i \omega_i - 1 = 0 \\
&\iff \sum_{i=1}^{N-1} \omega_i = 1 \\
\frac{\partial \mathcal{L}}{\partial \omega_i} &= \sum_{t=1}^T \left\{ (Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it}) \cdot 2 \cdot (-Y_{it}) + 2\tau W_{Nt} Y_{it} \right\} - \lambda_1 - \lambda_{2_i} = 0 \\
&\iff \sum_{t=1}^T -2Y_{it}(Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it} - \tau W_{Nt}) - \lambda_1 - \lambda_{2_i} = 0 \\
\frac{\partial \mathcal{L}}{\partial \tau} &= \sum_{t=1}^T \left\{ -2W_{Nt}(Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it}) + 2\tau W_{Nt} \right\} = 0 \\
&\iff \sum_{t=1}^T W_{Nt}(\tau - (Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it})) = 0 \\
&\iff \tau - (Y_{NT} - \sum_{i=1}^{N-1} \omega_i Y_{iT}) = 0 \quad \text{since } W_{Nt} \text{ is all zeros except a single 1 for } W_{NT} \\
&\iff \tau = (Y_{NT} - \sum_{i=1}^{N-1} \omega_i Y_{iT})
\end{aligned}$$

Thus, combining our expression for τ and the second FOC by recognizing that $W_{Nt} = 0 \forall t \neq T$, we find

$$\begin{aligned}
&\sum_{t=1}^{T-1} -2Y_{it}(Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it}) + \underbrace{(-2Y_{iT})(Y_{NT} - \sum_{i=1}^{N-1} \omega_i Y_{iT} - \tau)}_{=\tau} - \lambda_1 - \lambda_{2_i} = 0 \\
&\iff \sum_{t=1}^{T-1} -2Y_{it}(Y_{Nt} - \sum_{i=1}^{N-1} \omega_i Y_{it}) - \lambda_1 - \lambda_{2_i} = 0
\end{aligned}$$

We recognize that this is exactly the same FOC as for the SC problem. Thus, both optimization problems will yield the same answer and we can rewrite the ADH estimator as the desired regression estimator.

(c) We can get placebo variances for our estimator using either unit or time placebo studies. The general idea is that we pretend that either a unit or a time period received treatment that didn't in the actual application, we use the SC estimator to estimate the treatment effect and then lastly compare the predicted value to the true observed value. If the synthetic control weights are credible, we should be able to predict the "unobserved" value pretty well, as we should get a precise treatment effect of $\tau = 0$. Thus, we can use the difference between the actually observed outcome

and the predicted outcome as a measure for our error term in the model and use these error terms to get an idea for the variance of the estimator. If the variance is small, it supports the credibility of the SC estimator.